

COMBINATORICS

**Module 1: The Illusion of Certainty **

1. THE FUNDAMENTAL COUNTING PRINCIPLE

Before we can calculate probabilities, we have to know exactly how many possible realities exist. If an event is made up of multiple independent choices, you find the total number of outcomes by multiplying the number of options for each choice together.

- **The Rule:** If you have a ways to do one thing, b ways to do another, and c ways to do a third, the total number of combinations is $a \times b \times c$.
- **Example:** A digital safe requires a 3-digit PIN using the numbers 0 through 9.

Since there are 10 possible numbers for each slot, the math is: $10 \times 10 \times 10 = 1,000$ unique realities.

2. FACTORIALS: THE MATH OF EXPLOSION ($n!$)

When you want to arrange an entire set of distinct items into a specific order, you use a factorial. A factorial (represented by an exclamation mark) means multiplying a number by every whole number below it down to 1.

- **The Formula:** $n! = n \times (n - 1) \times (n - 2) \times \dots \times 1$
- **The 52-Card Reality Check:** The number of ways to arrange a standard deck of 52 cards is $52!$. This number (8.06×10^{67}) is so massive that every time you properly shuffle a deck, you are holding a specific sequence of cards that has never existed before in the history of the universe.

3. PERMUTATIONS: WHEN ORDER MATTERS

A permutation is a specific arrangement of items where **the order matters**. Think of a digital password, a padlock combination, or a race where coming in 1st, 2nd, or 3rd place makes a huge difference.

- **The Formula:** To find the number of ways to arrange k items chosen from a total pool of n items, use:

$$P(n, k) = \frac{n!}{(n - k)!}$$

- **Example:** If 8 horses are running a race, how many different ways can they finish in 1st, 2nd, and 3rd place?

$$n = 8 \text{ and } k = 3.$$

$$P(8, 3) = \frac{8!}{(8-3)!} = 336 \text{ possible podiums.}$$

4. COMBINATIONS: WHEN ORDER IS IRRELEVANT

A combination is a grouping of items where **the order does not matter**. Think of a hand of poker or choosing a team of people. Getting dealt an Ace then a King is the exact same hand as getting dealt a King then an Ace.

- **The Formula:** Because order doesn't matter, we have to divide out all the redundant permutations to avoid double-counting. We use the "n choose k" formula:

$$C(n, k) = \binom{n}{k} = \frac{n!}{k!(n - k)!}$$

- **Example:** If you want to pick a 3-person committee from a group of 8 people, how many unique teams can you make?

$$n = 8 \text{ and } k = 3.$$

$$C(8, 3) = \frac{8!}{3!(8-3)!} = 56 \text{ unique teams.}$$